Agile Software Development and DevOps Lab

Experiment 5

Aim: To automate application deployment using GitHub/Jenkins.

LO mapped: LO3

LO: Build and manage automated CI/CD pipelines for application development, testing, packaging and deployment.

Theory:

Continuous Integration (CI) and Continuous Deployment (CD) form a fundamental DevOps practice aimed at bridging the gap between development and operational activities.

- Continuous Integration (CI): A software development practice where developers regularly merge code changes into a central repository, followed by automated builds and automated tests. This helps catch bugs early, increases software quality, and reduces validation overhead.
- Continuous Deployment (CD): The automated process where verified changes in the codebase are automatically packaged, configured, and deployed directly to production or staging runtime environments without manual intervention.
- Jenkins & Declarative Pipelines: Jenkins is an open-source automation server enabling automated build, test, and deployment workflows. A Declarative Pipeline uses structured Groovy-based domain-specific language (DSL) to define stages, steps, and build environments as code (Jenkinsfile), making infrastructure version-controlled and reproducible.

Step 1: Jenkins Server Initialization and Access Setup

- Jenkins runs as a background Windows service requiring a compatible Java Development Kit (JDK 21) runtime environment to execute its core processes.

- Accessing the web console at http://localhost:8080 allows the setup wizard to be completed, during which the first administrator user is created to manage permissions, job configurations and server tasks.

Step 2: Baseline Pipeline Validation (Inline Scripting)

- A new project item named GitHub-Jenkins-Demo is created using the native Pipeline job type.
- Before linking to external version control, an inline Declarative Pipeline script is added directly to the job configuration to test the server's execution engine.
- The script utilizes the agent any directive along with three core lifecycle stages: Build, Test, and Deploy, each executing standard echo statements.
- Running Build #1 causes Jenkins to assign a local node executor in the workspace directory, process each stage sequentially, and display the console logs showing that the pipeline completed with SUCCESS.

Step 3: Source Code Management (SCM) & Pipeline-as-Code Implementation

- A central GitHub repository named jenkins-demo-app is established under user profile to maintain the application codebase and its automation configuration.
- An index.html file is added as the frontend application artifact containing an initial header message.
- A version-controlled Jenkinsfile is defined in the root directory, following Declarative Pipeline syntax.
- In the Deploy stage of the Jenkinsfile, automated batch instructions (bat) are scripted to verify the destination directory C:\Deployments\DemoApp and copy index.html from the workspace to the target folder.

Step 4: Connecting Jenkins to GitHub and Initial Deployment

- The Jenkins project configuration is transitioned from an inline script to Pipeline script from SCM, specifying Git as the source provider.
- Jenkins is supplied with the repository HTTPS URL, the branch specifier is configured to */main, and the script path is set to target the repository's Jenkinsfile.
- Executing Build #2 prompts Jenkins to clone the remote repository, check out the latest commit, interpret the Jenkinsfile, and run the deployment batch script.

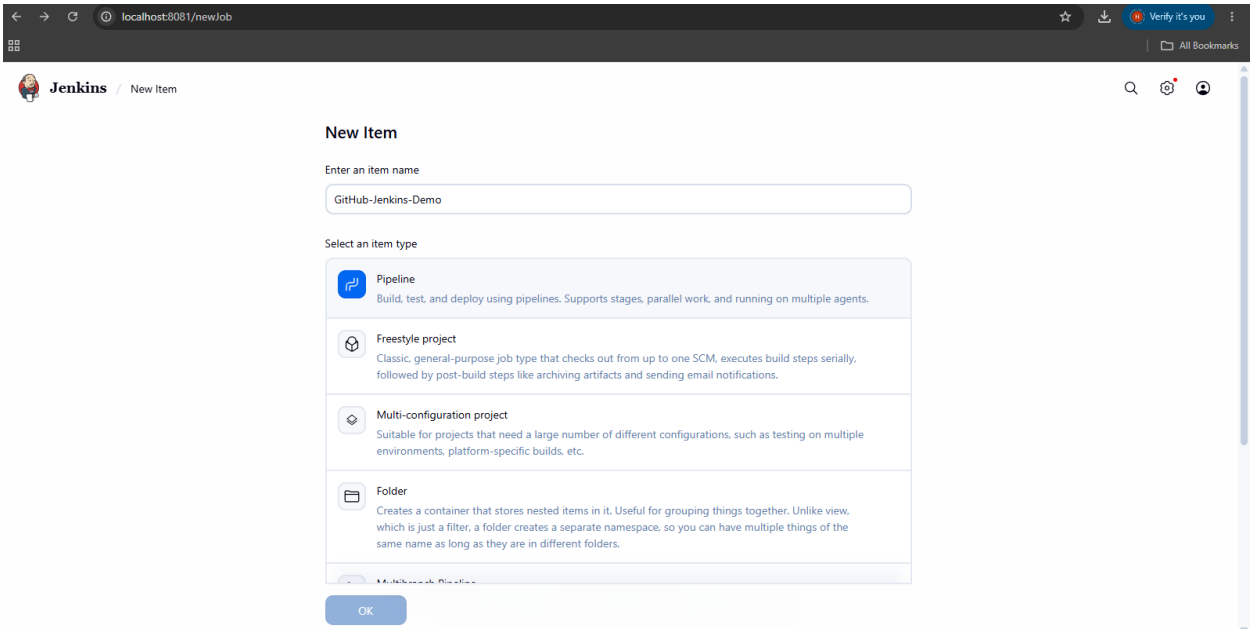
- Opening C:/Deployments/DemoApp/index.html in a web browser verifies that the initial frontend file was deployed successfully.

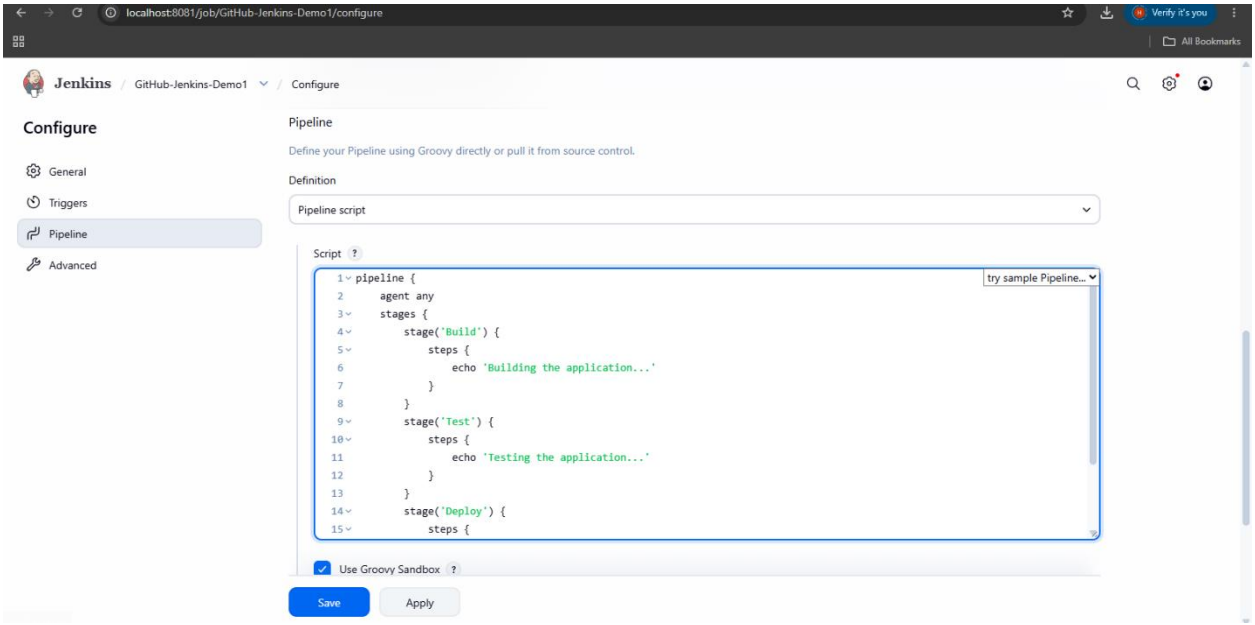
Step 5: Continuous Integration & Continuous Deployment (CI/CD) Verification

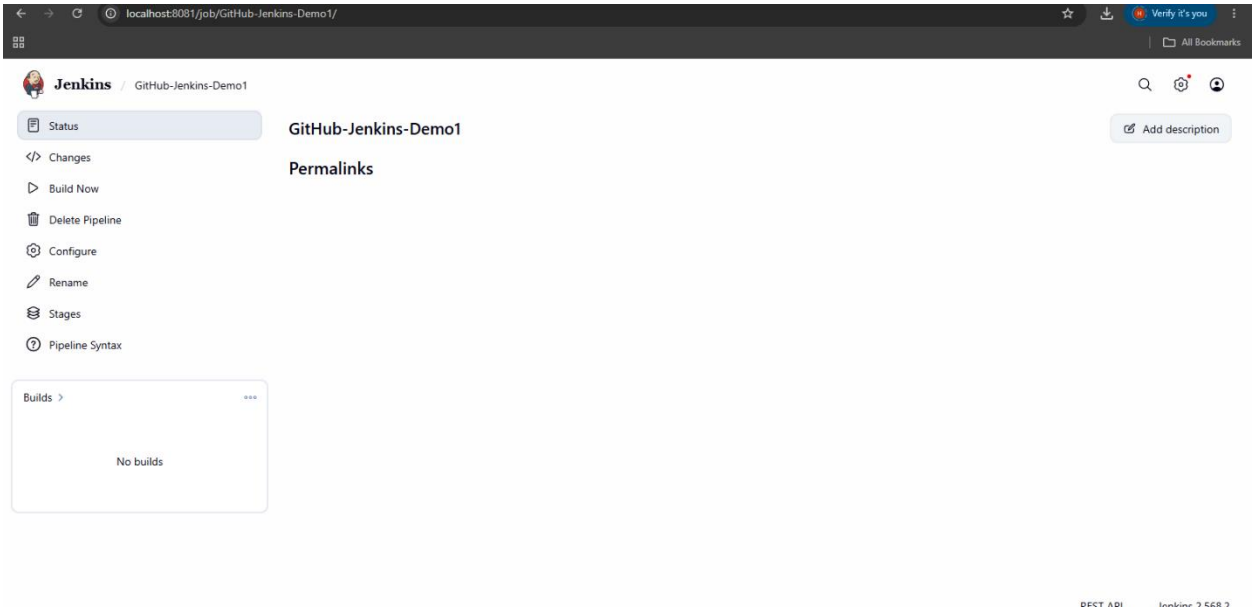
- To simulate real-world continuous deployment, a code change is made directly in index.html on GitHub by adding custom CSS styling and modifying the text to Application Updated & Deployed via CI/CD Pipeline! Version 2.0 is live..
- Upon running Build #3, Jenkins pulls the updated commit hash, executes the stages, and overwrites the existing index.html in C:\Deployments\DemoApp with the new version.
- Refreshing the browser page confirms that the new blue-themed card UI and Version

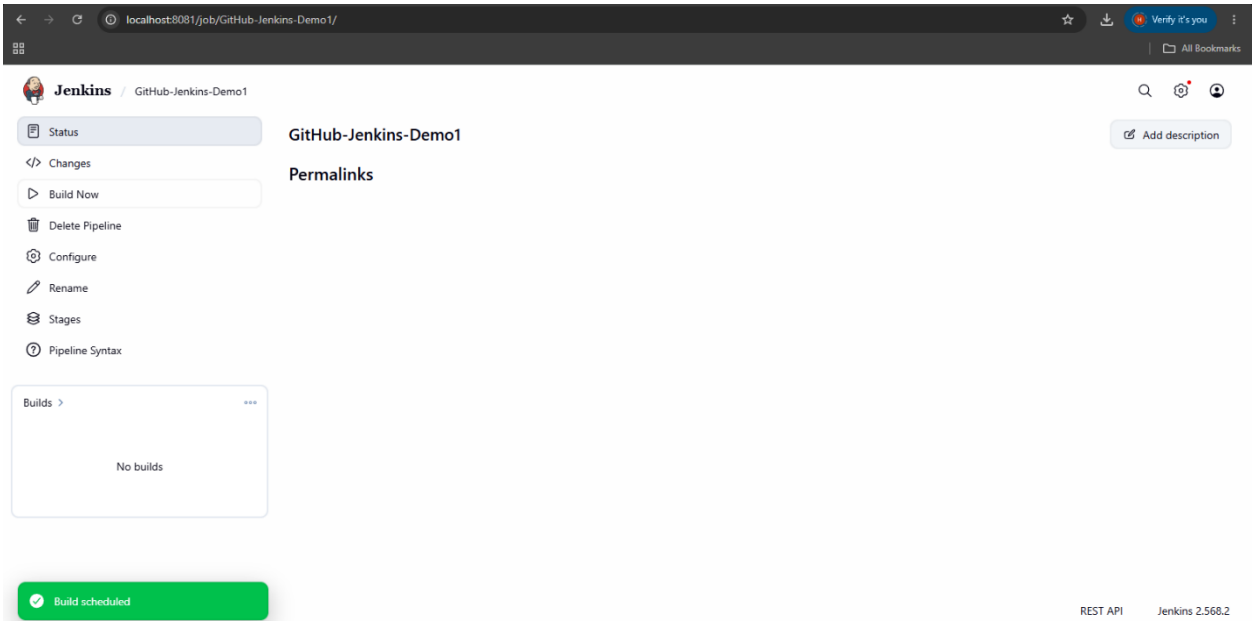
2.0 text are rendered, validating the full automated delivery pipeline from source control to local target deployment.

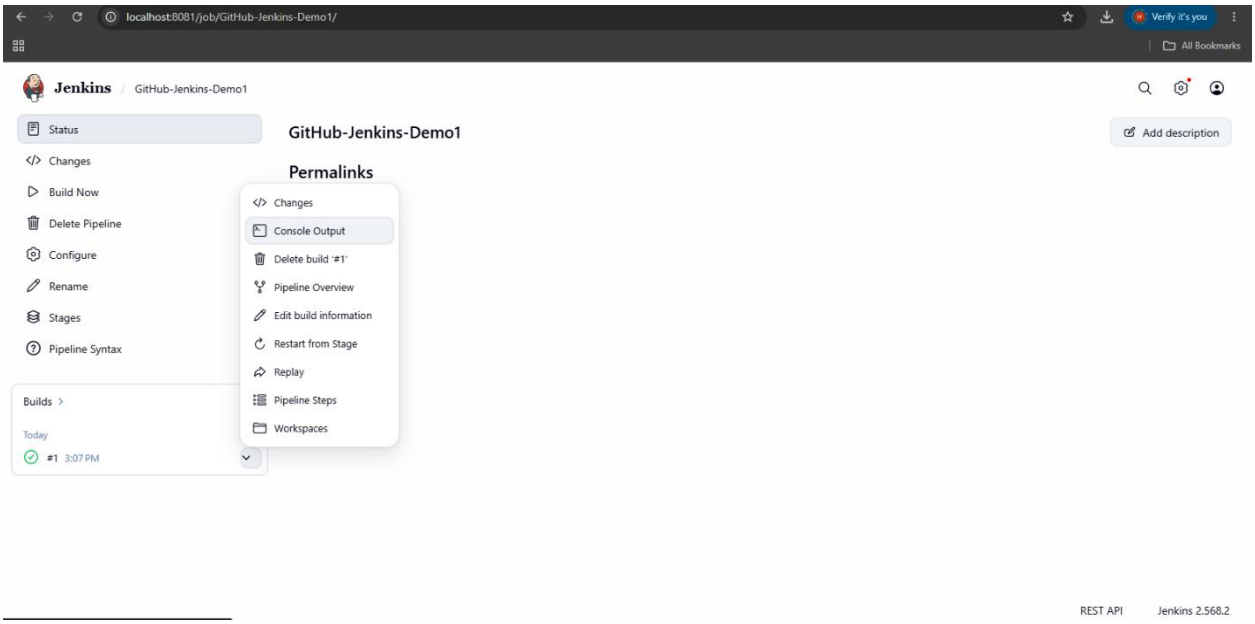
Output:

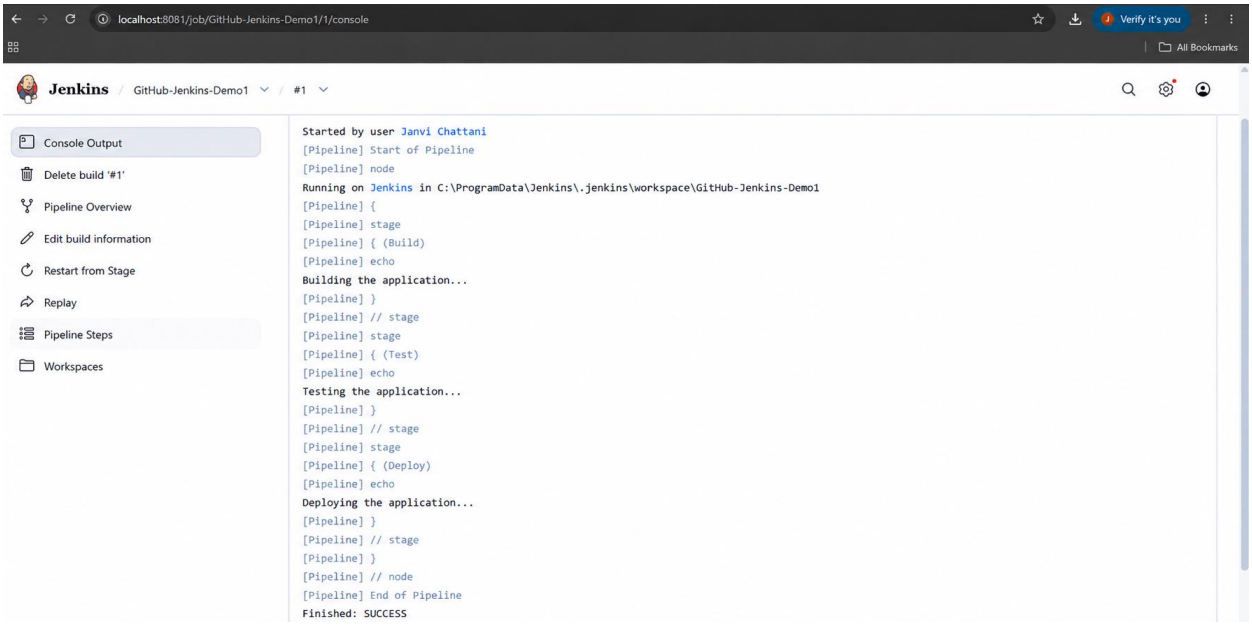


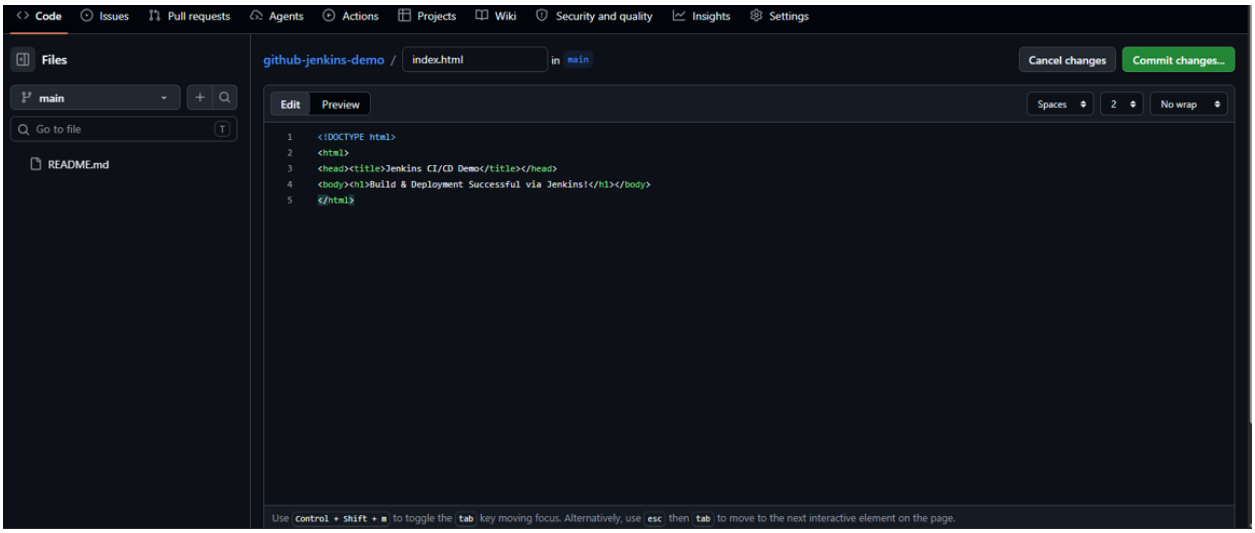

localhost:8081/job/GitHub-Jenkins-Demo1/configure
Verify it's you
All Bookmarks
Jenkins / GitHub-Jenkins-Demo1 / Configure
Configure
General
Triggers
Pipeline
Advanced
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script ?
try sample Pipeline...
pipeline {
agent any
stages {
stage('Build') {
steps {
echo 'Building the application...'
}
}
stage('Test') {
steps {
echo 'Testing the application...'
}
}
stage('Deploy') {
steps {
Use Groovy Sandbox ?
Save
Apply

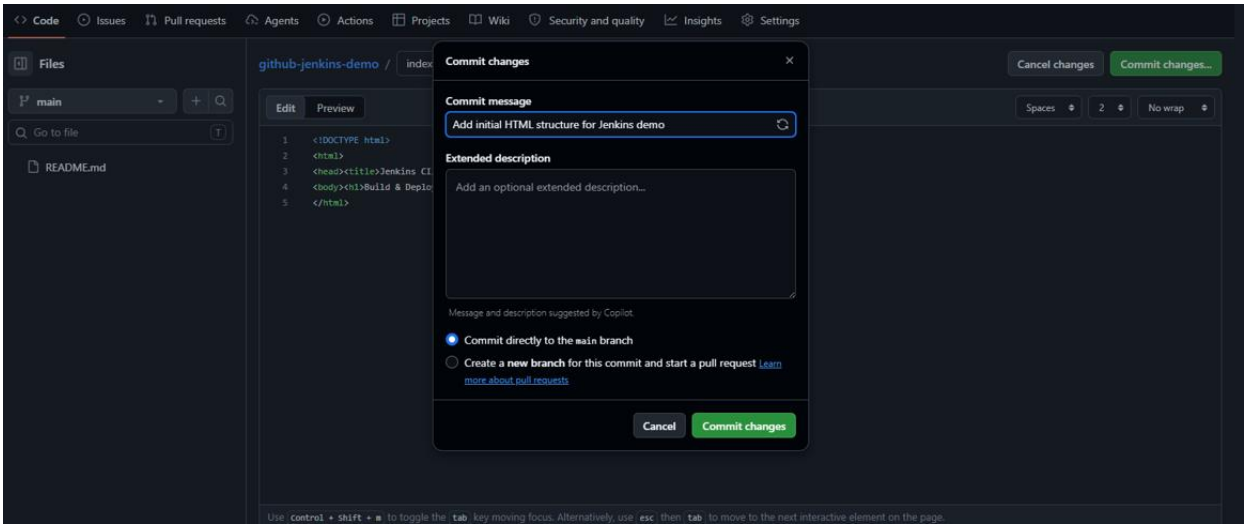

localhost:8081/job/GitHub-Jenkins-Demo1/
Verify it's you
All Bookmarks
Jenkins / GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
Builds
No builds
GitHub-Jenkins-Demo1
Add description
Permalinks
REST API

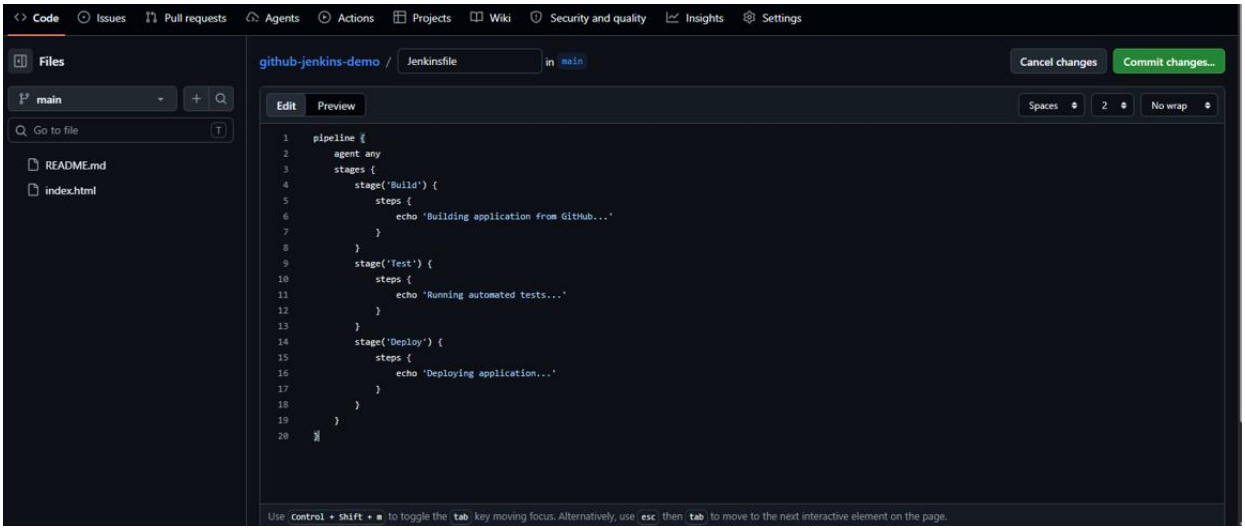

localhost:8081/job/GitHub-Jenkins-Demo1/
Verify it's you
All Bookmarks
Jenkins
GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub-Jenkins-Demo1
Add description
Permalinks
Builds
No builds
Build scheduled
REST API
Jenkins 2.568.2

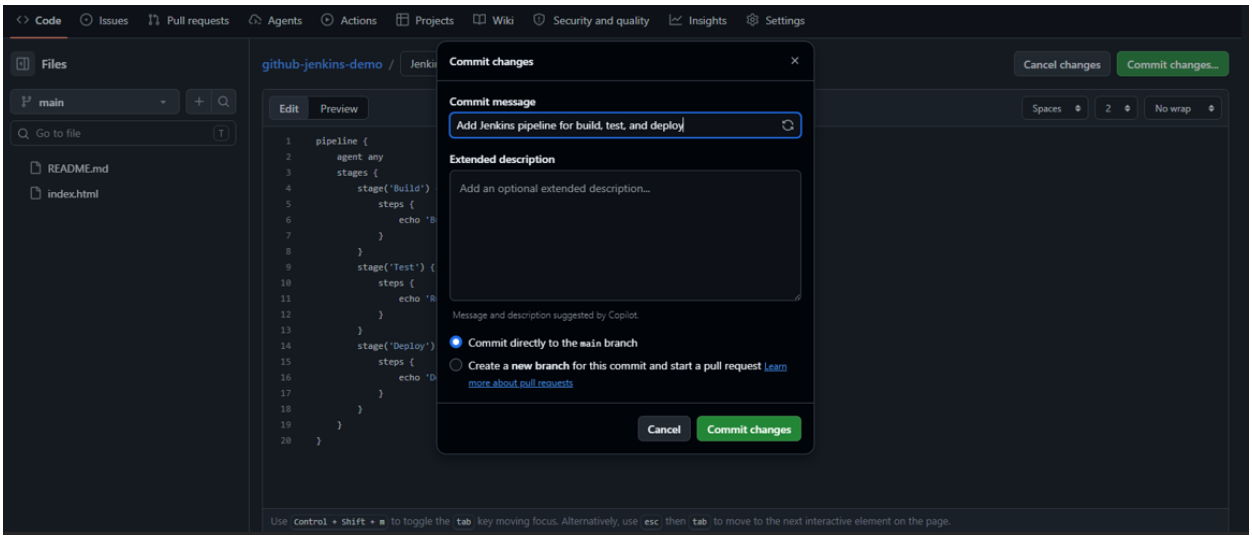

localhost:8081/job/GitHub-Jenkins-Demo1/
Verify it's you
All Bookmarks
Jenkins
GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub-Jenkins-Demo1
Add description
Permalinks
Changes
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Builds
Today
#1 3:07 PM
REST API
Jenkins 2.568.2


localhost:8081/job/GitHub-Jenkins-Demo1/1/console
Verify it's you
All Bookmarks
Jenkins
GitHub-Jenkins-Demo1
#1
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Started by user Janvi Chattani
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo1
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Testing the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

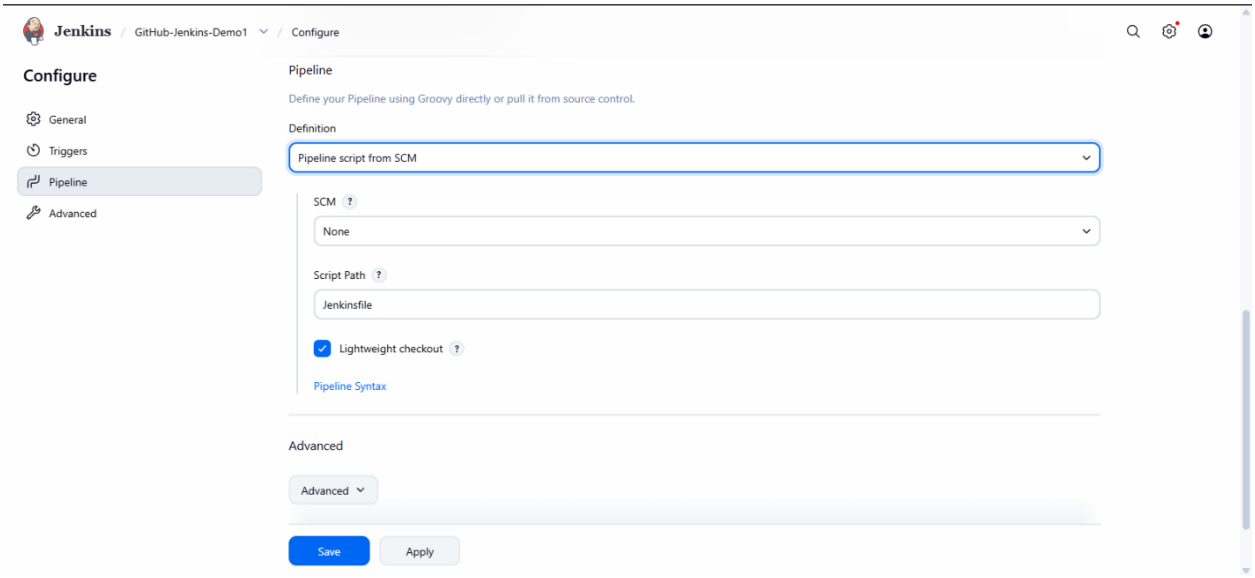

Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Files
main
Go to file
README.md
github-jenkins-demo / index.html in main
Cancel changes
Commit changes...
Edit
Preview
Spaces
2
No wrap
<!DOCTYPE html>
<html>
<head><title>Jenkins CI/CD Demo</title></head>
<body><h1>Build & Deployment Successful via Jenkins!</h1></body>
</html>
Use Control + Shift + m to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element on the page.

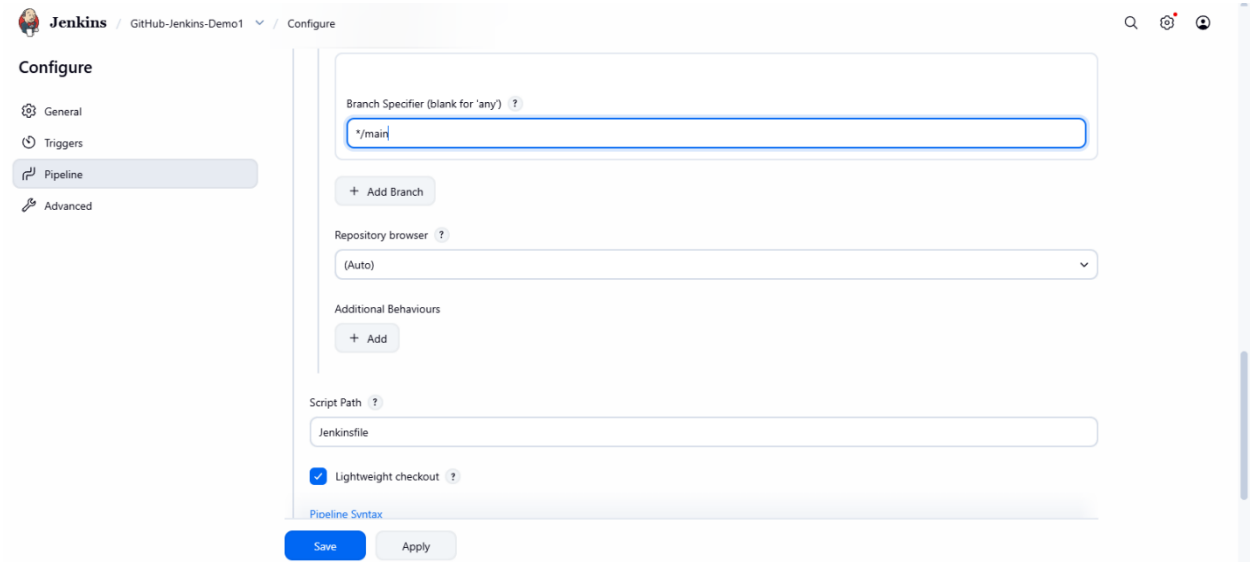

Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Files
main
Go to file
README.md
github-jenkins-demo
Cancel changes
Commit changes...
Edit
Preview
Commit changes
Commit message
Add initial HTML structure for Jenkins demo
Extended description
Add an optional extended description...
Message and description suggested by Copilot.
Commit directly to the main branch
Create a new branch for this commit and start a pull request Learn more about pull requests
Cancel
Commit changes

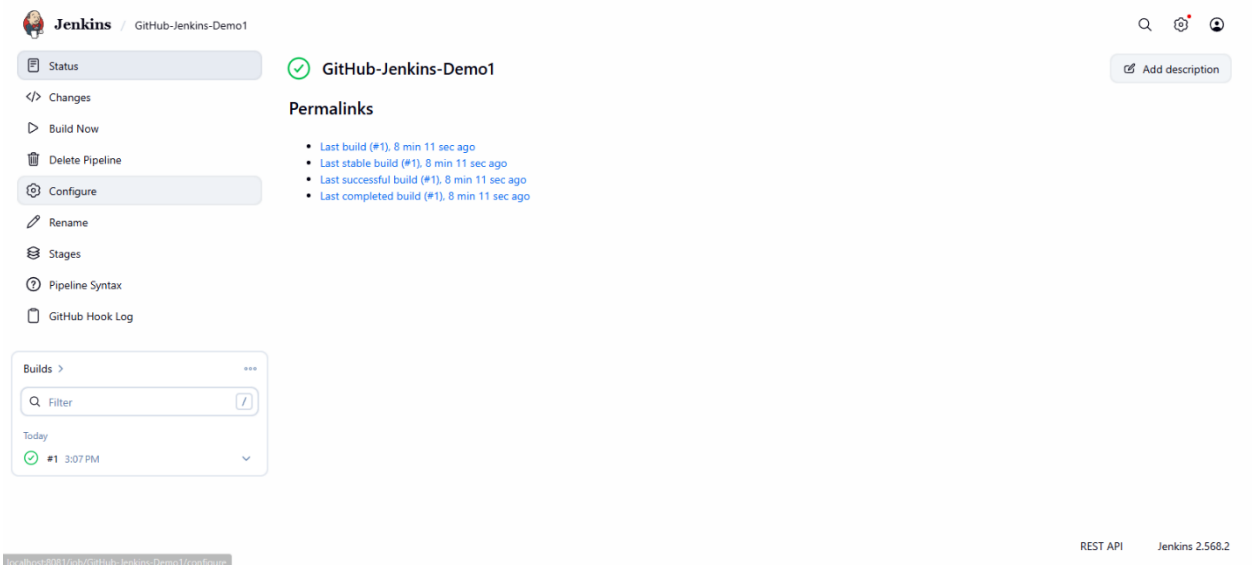

Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Files
main
Go to file
README.md
index.html
github-jenkins-demo / Jenkinsfile in main
Cancel changes
Commit changes...
Edit
Preview
Spaces
2
No wrap
pipeline {
agent any
stages {
stage('Build') {
steps {
echo 'Building application from GitHub...'
}
}
stage('Test') {
steps {
echo 'Running automated tests...'
}
}
stage('Deploy') {
steps {
echo 'Deploying application...'
}
}
}
}
Use Control + Shift + m to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element on the page.

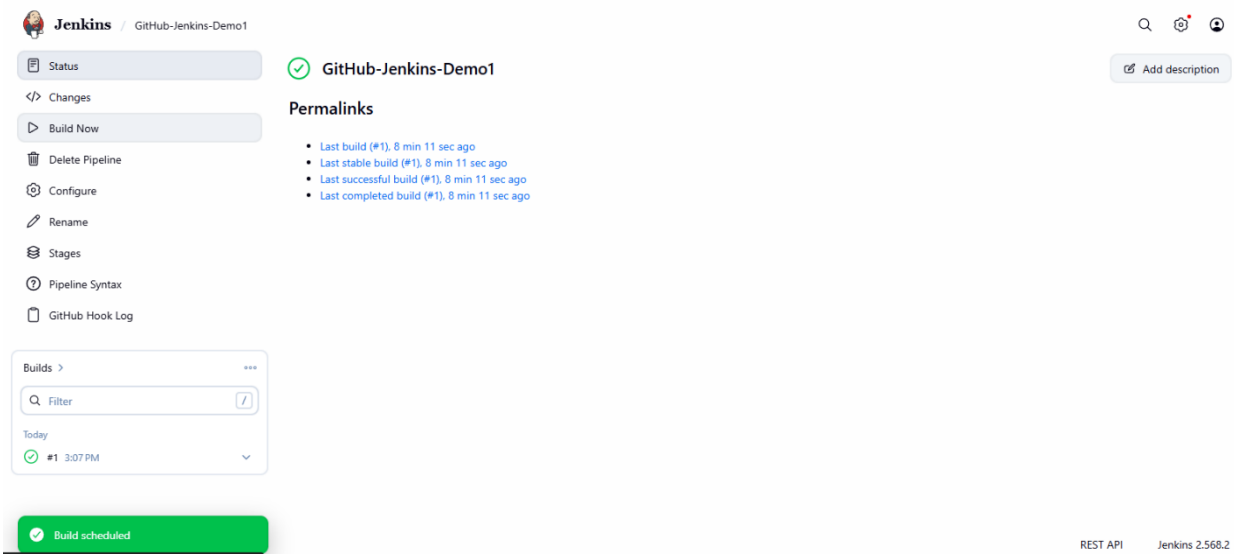

Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Files
main
Go to file
README.md
index.html
github-jenkins-demo /
Cancel changes
Commit changes...
Edit
Preview
Spaces
2
No wrap
Commit changes
Commit message
Add Jenkins pipeline for build, test, and deploy
Extended description
Add an optional extended description...
Message and description suggested by Copilot.
Commit directly to the main branch
Create a new branch for this commit and start a pull request Learn more about pull requests
Cancel
Commit changes
Use Control + Shift + m to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element on the page.

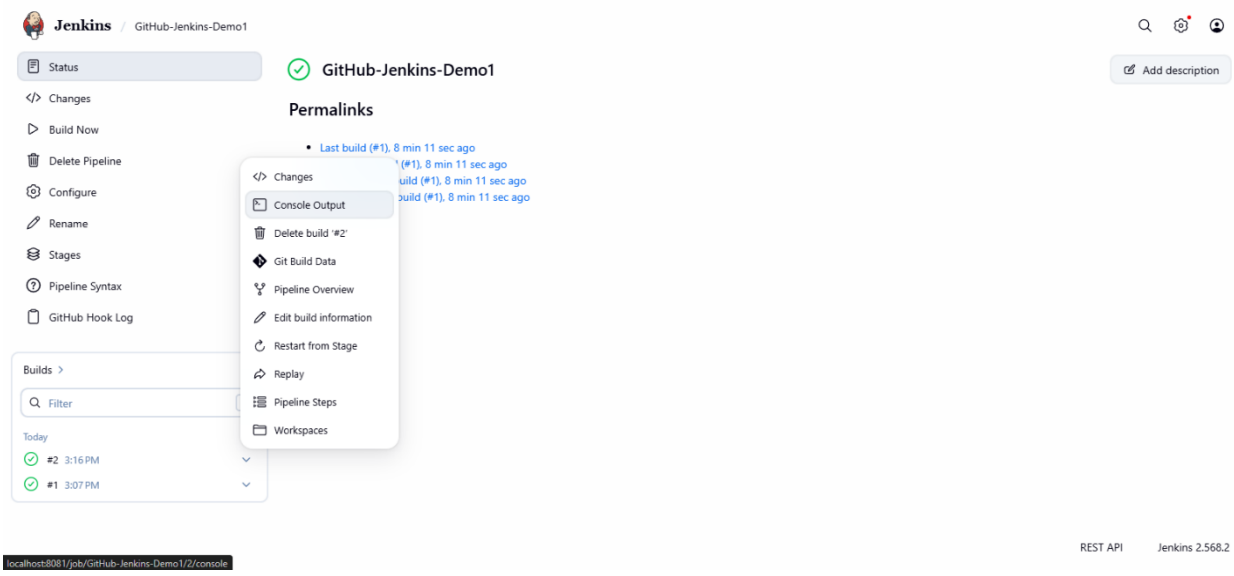

Jenkins
GitHub-Jenkins-Demo1
Configure
Configure
General
Triggers
Pipeline
Advanced
Triggers
Set up automated actions that start your build based on specific events, like code changes or scheduled times.
Build after other projects are built
Build periodically
GitHub hook trigger for GITScm polling
Poll SCM
Trigger builds remotely (e.g., from scripts)
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script
pipeline {
agent any
stages {
stage('Build') {
Save
Apply

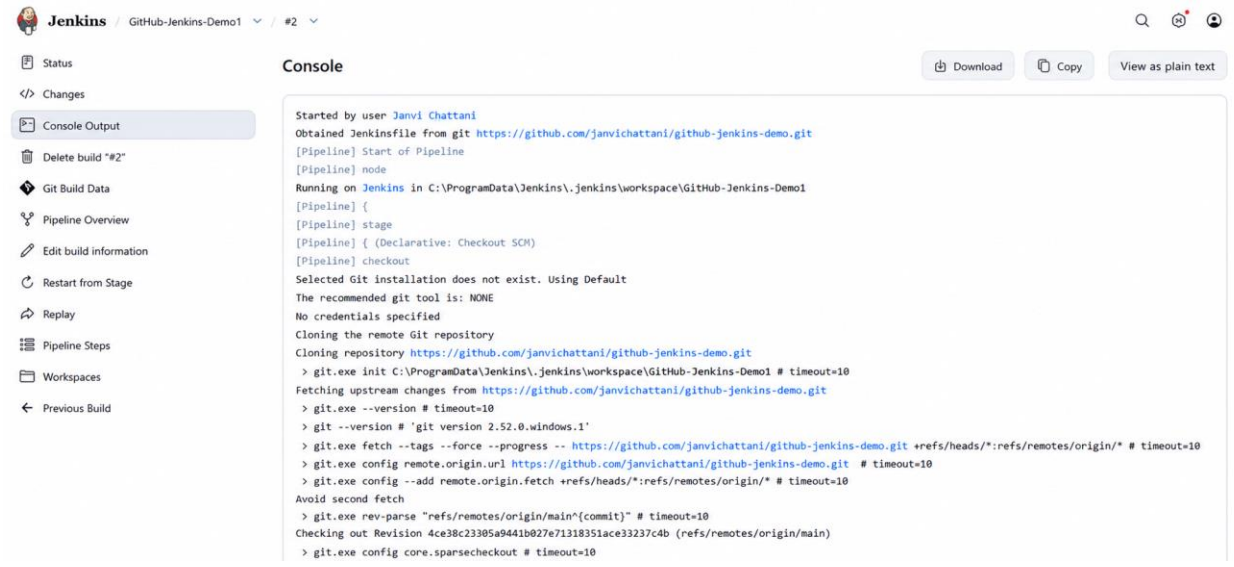

Jenkins / GitHub-Jenkins-Demo1 / Configure
Configure
General
Triggers
Pipeline
Advanced
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script from SCM
SCM
None
Script Path
Jenkinsfile
Lightweight checkout
Pipeline Syntax
Advanced
Advanced
Save
Apply


Jenkins / GitHub-Jenkins-Demo1 / Configure
Configure
General
Triggers
Pipeline
Advanced
Branch Specifier (blank for 'any')
*/main
Add Branch
Repository browser
(Auto)
Additional Behaviours
Add
Script Path
Jenkinsfile
Lightweight checkout
Pipeline Syntax
Save
Apply


Jenkins / GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub Hook Log
Builds
Filter
Today
#1 3:07 PM
GitHub-Jenkins-Demo1
Add description
Permalinks
Last build (#1), 8 min 11 sec ago
Last stable build (#1), 8 min 11 sec ago
Last successful build (#1), 8 min 11 sec ago
Last completed build (#1), 8 min 11 sec ago
REST API
Jenkins 2.568.2


Jenkins / GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub Hook Log
Builds
Filter
Today
#1 3:07 PM
Build scheduled
GitHub-Jenkins-Demo1
Add description
Permalinks
Last build (#1), 8 min 11 sec ago
Last stable build (#1), 8 min 11 sec ago
Last successful build (#1), 8 min 11 sec ago
Last completed build (#1), 8 min 11 sec ago
REST API
Jenkins 2.568.2


Jenkins / GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub Hook Log
Builds
Filter
Today
#2 3:16 PM
#1 3:07 PM
Changes
Console Output
Delete build '#2'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
GitHub-Jenkins-Demo1
Add description
Permalinks
Last build (#1), 8 min 11 sec ago
REST API
Jenkins 2.568.2
localhost:8081/job/GitHub-Jenkins-Demo1/2/console


Jenkins / GitHub-Jenkins-Demo1 / #2
Status
Changes
Console Output
Delete build "#2"
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Console
Download
Copy
View as plain text
Started by user Janvi Chattani
Obtained Jenkinsfile from git https://github.com/janvichattani/github-jenkins-demo.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo1
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/janvichattani/github-jenkins-demo.git
> git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo1 # timeout=10
Fetching upstream changes from https://github.com/janvichattani/github-jenkins-demo.git
> git.exe --version # timeout=10
> git --version # 'git version 2.52.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/janvichattani/github-jenkins-demo.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe config remote.origin.url https://github.com/janvichattani/github-jenkins-demo.git # timeout=10
> git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 4ce38c23305a9441b027e71318351ace33237c4b (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10

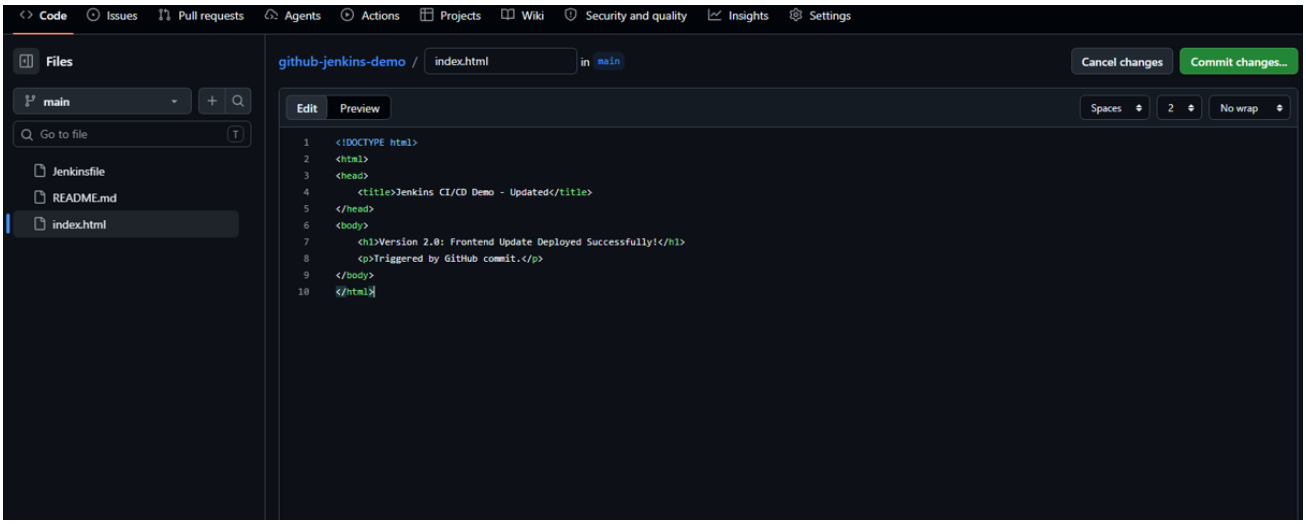
Jenkins / GitHub-Jenkins-Demo1 / #2

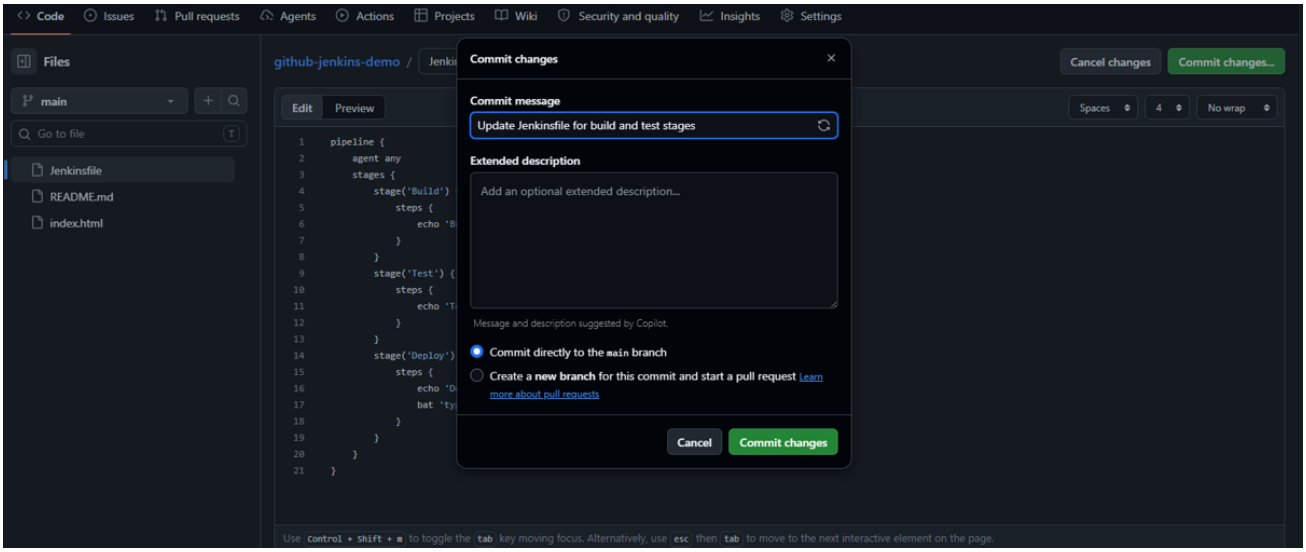
```
Checking out Revision 4ce38c23305a9441b027e71318351ace33237c4b (refs/remotes/origin/main)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 4ce38c23305a9441b027e71318351ace33237c4b # timeout=10
Commit message: "Add Jenkins pipeline for build, test, and deploy"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building application from GitHub...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running automated tests...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
```

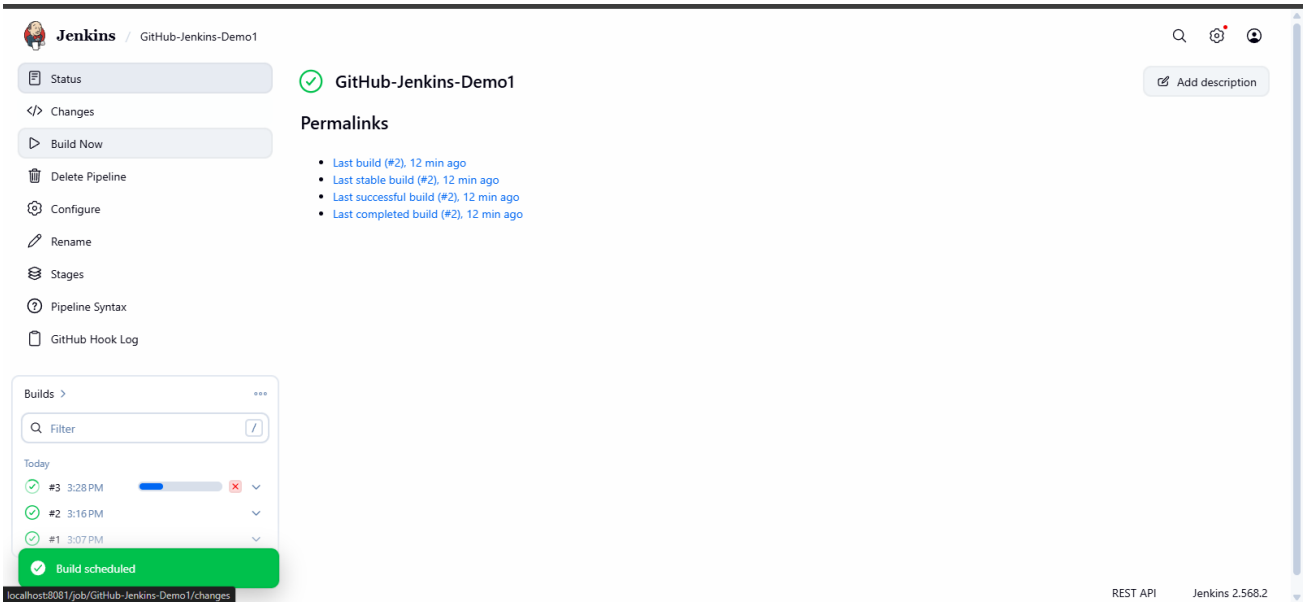
Jenkins / GitHub-Jenkins-Demo1 / #2

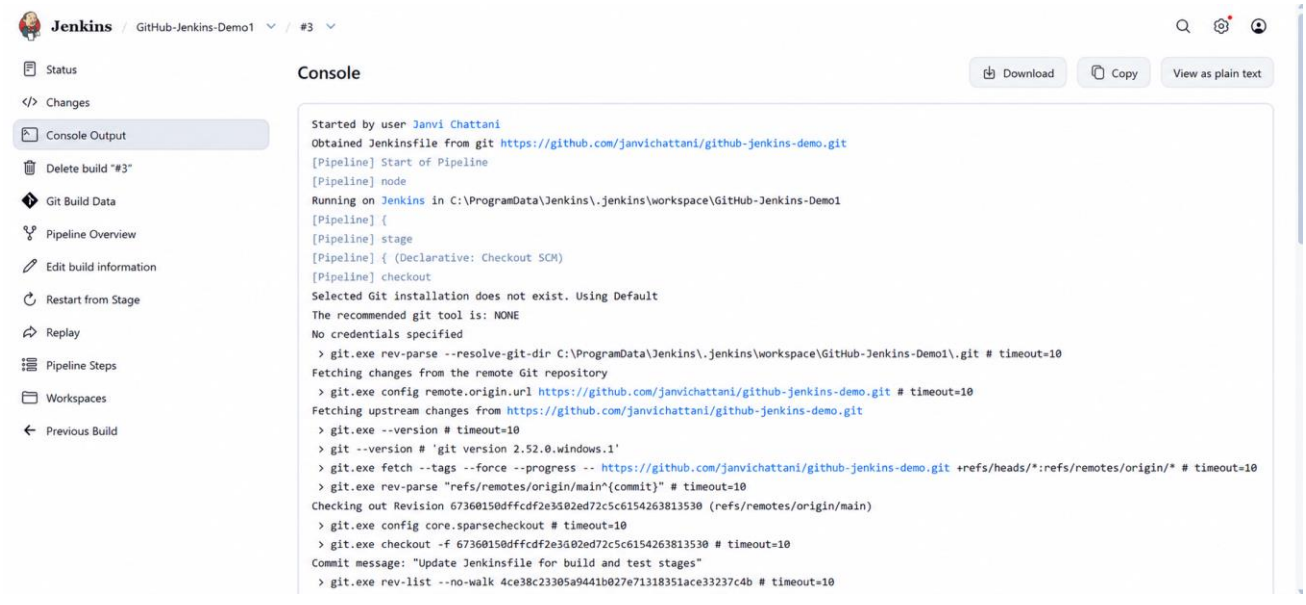
```
[Pipeline] { (Build)
[Pipeline] echo
Building application from GitHub...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running automated tests...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.568.2




Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Files
main
Go to file
Jenkinsfile
README.md
index.html
github-jenkins-demo
Commit changes
Commit message
Update Jenkinsfile for build and test stages
Extended description
Add an optional extended description...
Message and description suggested by Copilot.
Commit directly to the main branch
Create a new branch for this commit and start a pull request Learn more about pull requests
Cancel
Commit changes
Cancel changes
Commit changes...
Edit
Preview
Spaces
4
No wrap
pipeline {
agent any
stages {
stage('Build')
steps {
}
stage('Test') {
steps {
}
stage('Deploy')
steps {
bat 'ty
}
}
}


Jenkins
GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub Hook Log
GitHub-Jenkins-Demo1
Add description
Permalinks
Last build (#2), 12 min ago
Last stable build (#2), 12 min ago
Last successful build (#2), 12 min ago
Last completed build (#2), 12 min ago
Builds
Filter
Today
#3 3:28 PM
#2 3:16 PM
#1 3:07 PM
Build scheduled
localhost:8081/job/GitHub-Jenkins-Demo1/changes
REST API
Jenkins 2.568.2


Jenkins
GitHub-Jenkins-Demo1
#3
Status
Changes
Console Output
Delete build "#3"
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Console
Download
Copy
View as plain text
Started by user Janvi Chattani
Obtained Jenkinsfile from git https://github.com/janvichattani/github-jenkins-demo.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo1
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo1\.git # timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/janvichattani/github-jenkins-demo.git # timeout=10
Fetching upstream changes from https://github.com/janvichattani/github-jenkins-demo.git
> git.exe --version # timeout=10
> git --version # 'git version 2.52.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/janvichattani/github-jenkins-demo.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 67360150dffcdf2e3302ed72c5c6154263813530 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 67360150dffcdf2e3302ed72c5c6154263813530 # timeout=10
Commit message: "Update Jenkinsfile for build and test stages"
> git.exe rev-list --no-walk 4ce38c23305a9441b027e71318351ace33237c4b # timeout=10





Version 2.0: Frontend Update Deployed Successfully!

Triggered by GitHub commit.